Mid-Term Project: Reflectivity-Augmented LiDAR Scene Understanding for Robotic Perception

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***Abstract*—This mid-term project studies whether a lightweight range-aware transformation of LiDAR intensity can provide a useful complementary cue for semantic scene understanding. The work is motivated by the observation that raw LiDAR intensity is not a pure surface descriptor, because it is entangled with measurement geometry, especially range, and more broadly with incidence effects. Rather than claiming calibrated reflectivity recovery, this preliminary study adopts a disciplined proxy formulation, $\hat{\rho}_i = I_i \cdot R_i$, where I_i is raw intensity and R_i is point-wise Euclidean range. The project uses SemanticKITTI sequence 00 and is organized as a staged notebook pipeline covering data verification, single-frame signal analysis, short-window motion analysis, single-frame semantic analysis, and multi-frame semantic consistency. Results show that the pseudo-reflectivity proxy is not merely a trivial rescaling of intensity, but changes signal distributions, class ordering, and spatial emphasis in meaningful ways. On the semantic frame used in the study, the multiclass separability score improved from 0.4482 for raw intensity to 0.5055 for pseudo-reflectivity. Across a short 10-frame semantic consistency window, class-ordering structure remained highly stable for both signals, with a mild average advantage for the proxy. Overall, the preliminary evidence suggests that a simple range-aware pseudo-reflectivity proxy can yield modest but meaningful gains in semantic interpretability, while remaining a heuristic, range-dependent feature rather than true calibrated reflectivity.**

I. Introduction

LiDAR sensors provide robotic systems with sparse but geometrically meaningful measurements of the surrounding environment. Each point carries spatial coordinates and a recorded return strength, commonly referred to as intensity. While intensity can contain useful information, it is not directly equivalent to reflectivity or material identity, because it is affected by range and, more broadly, by measurement geometry such as incidence effects. This makes raw intensity difficult to interpret as a stable surface cue.

This project was inspired by recent research arguing that reflectivity-aware reasoning can provide a more consistent and interpretable signal for semantic LiDAR understanding. However, full reflectivity calibration is outside the scope of the current mid-term stage. The goal here is therefore deliberately narrower: to test whether a simple range-aware pseudo-reflectivity proxy can expose more useful semantic structure than raw intensity alone.

The present work is not a segmentation-training paper and does not claim material identification. Its purpose is to establish feasibility, verify the data pipeline, inspect the behavior of the proxy under short-window motion, and determine whether the idea is strong enough to justify a broader final-stage study.

II. Concepts and Mathematical Model

Each LiDAR point is represented as

$$p_i = (x_i, y_i, z_i, I_i), \tag{1}$$

where (x_i, y_i, z_i) are spatial coordinates and I_i is raw LiDAR intensity.

For each point, Euclidean range is computed as

$$R_i = \sqrt{x_i^2 + y_i^2 + z_i^2}. \tag{2}$$

A simplified LiDAR return interpretation is

$$I_i \propto \frac{\rho_i \cos(\alpha_i)}{R_i^2}, \tag{3}$$

where ρ_i is surface reflectivity, α_i is incidence angle, and R_i is range. This relation motivates the project's central idea: raw intensity is informative, but it is not a clean intrinsic surface descriptor.

Because full calibration terms are outside the present scope, the preliminary notebooks use a practical pseudo-reflectivity proxy:

$$\hat{\rho}_i = I_i \cdot R_i. \tag{4}$$

This is not calibrated physical reflectivity. It is a lightweight range-aware feature whose value lies in semantic interpretability rather than exact reflectance recovery.

To evaluate semantic usefulness, the project uses a Fisher-style multiclass separability score based on between-class and within-class dispersion. For the motion-based semantic consistency analysis, class-order stability across frames is measured using Spearman rank correlation relative to an average class profile over the selected time window.

III. Simulation and Implementation Output

The preliminary work uses SemanticKITTI sequence 00 and is organized into five notebooks:

- **Notebook 01:** data loading and sanity checks,
- **Notebook 02:** single-frame reflectivity-style analysis,
- **Notebook 03:** multi-frame reflectivity-style analysis over motion,
- **Notebook 04:** single-frame semantic reflectivity analysis,

- **Notebook 05:** multi-frame semantic consistency over motion.

The main implementation outputs of the preliminary stage include:

- verified point-label alignment and structural sanity checks,
- BEV and forward-view visualizations of geometry, raw intensity, and pseudo-reflectivity,
- class-wise summary tables for semantic analysis,
- a 30-frame motion GIF comparing raw intensity and pseudo-reflectivity behavior,
- a labeled 30-frame reflectivity-augmented semantic GIF,
- compact temporal plots showing class-order consistency across motion.

Representative visual outputs are shown in Fig. 1 and Fig. 2. Figure 1 presents a representative single-frame comparison of bird's-eye-view and forward-view geometry, raw intensity, and pseudo-reflectivity, while Fig. 2 shows a reflectivity-augmented semantic visualization in which semantic identity is encoded by color and pseudo-reflectivity modulates brightness.

These outputs are designed to document the implemented pipeline, visualize the proxy's behavior, and provide early evidence for or against the project idea before moving to a broader final-stage contribution.

IV. Results

The preliminary results support the claim that the pseudo-reflectivity proxy is doing something meaningful, while also showing that its usefulness is not universal.

A. Single-Frame Signal Analysis

On the selected single frame, the proxy changed the signal statistics in a nontrivial way. The range-aware transformation altered the signal distribution, changed range dependence, and redistributed spatial emphasis relative to raw intensity.

B. Single-Frame Semantic Analysis

In the semantic analysis, the multiclass separability score improved from

$$0.4482 \quad \text{(raw intensity)} \tag{5}$$

to

$$0.5055 \quad \text{(pseudo-reflectivity).} \tag{6}$$

This corresponds to an absolute improvement of 0.0574 and a relative gain of about 12.8%.

C. Multi-Frame Semantic Consistency

Across a short 10-frame semantic consistency window, class ordering remained highly stable for both signals. The mean Spearman rank correlation relative to the mean class profile was:

$$0.9534 \quad \text{for raw intensity,} \tag{7}$$

$$0.9582 \quad \text{for pseudo-reflectivity.} \tag{8}$$

This indicates a mild average advantage for the proxy rather than a dramatic dominance claim.

V. Discussion and Potential Future Work

The preliminary evidence suggests that the proxy $I \cdot R$ is not merely a cosmetic rescaling of raw intensity. It changes class ordering, signal spread, and spatial emphasis in ways that can improve semantic interpretability. This is especially important because the project does not yet use a trained semantic segmentation model. The value of the current stage lies in disciplined signal analysis, not in downstream model performance.

At the same time, the results must be interpreted carefully. The proxy remains range-dependent and scene-dependent, and should not be presented as calibrated reflectivity. The observed gains are modest rather than transformative, and the short-window motion analysis shows that the advantage is not uniform across all frames or scene regimes. These limitations are not failures of the preliminary stage. They are exactly the clues that motivate the final phase: broader temporal evaluation, alternative proxy formulations, stronger metrics, explicit failure-case analysis, and lightweight downstream semantic testing.

Tentative final-stage extension plans include longer temporal studies over broader parts of the sequence, comparison of alternative range-aware proxy formulations beyond $I \cdot R$, stronger class-sensitive evaluation metrics, and more explicit analysis of frames or classes where the proxy helps only weakly or becomes disadvantageous. A further practical direction is to test whether pseudo-reflectivity provides useful complementary information in a lightweight downstream semantic task, while also packaging the current notebook workflow into a cleaner modular pipeline.

VI. Project Link

https://github.com/aymisxx/reflect-aug-seg

Acknowledgment

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References

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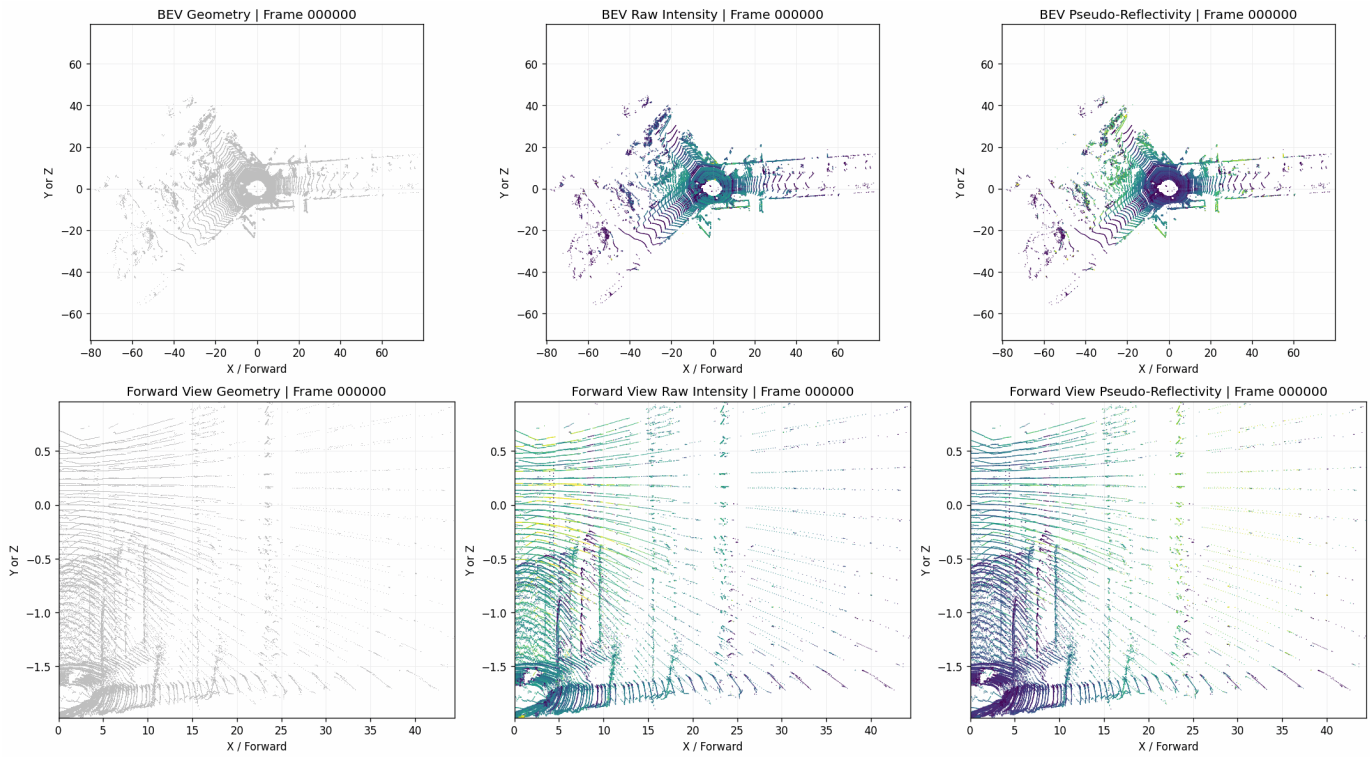

Fig. 1. Representative single-frame visualization from the preliminary pipeline showing bird's-eye-view and forward-view geometry, raw intensity, and pseudo-reflectivity for Frame 000000. The comparison illustrates that the proxy $\hat{\rho} = I \cdot R$ is not a trivial rescaling of intensity, but changes spatial emphasis across the scene.

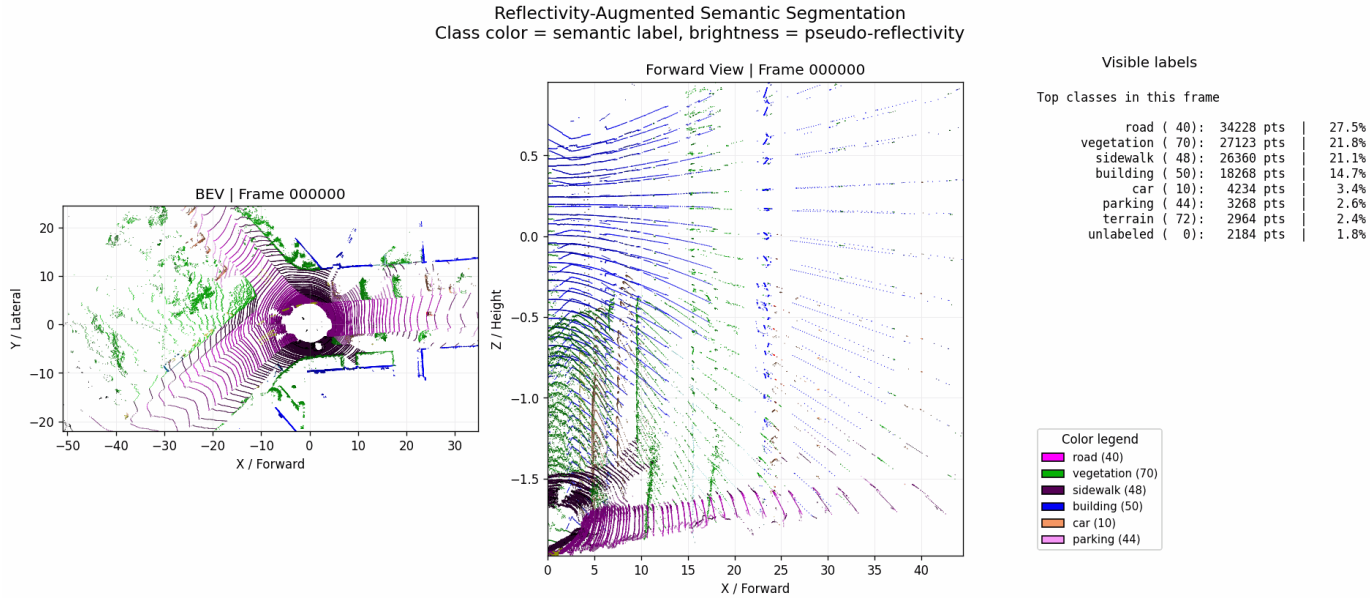

Fig. 2. Representative reflectivity-augmented semantic visualization for Frame 000000. Semantic class identity is encoded by color, while brightness is modulated by pseudo-reflectivity. This figure supports the single-frame semantic analysis and illustrates how semantic structure and range-aware signal variation can be viewed together.